

Assignment

Page No

Date

Day 2.

task 1. list 3 real world devices in your home or collage that use smps and 2 devices that use ups explaining in one line why each device needs that type of power supply.

→

* devices using smps :-

* Desktop PC :-

converts AC wall power into multiple stable, low DC voltages efficiently without generating excess heat.

* Smart TV :-

Requires a compact, lightweight power supply to step down high voltage to feed sensitive LED screens and microchips.

* Wi-Fi Router :-

Needs a continuous, regulated low-voltage DC power source in a small form factor to run 24/7 without overheating.

* Devices using ups (uninterruptible power supply) &

* Desktop computers :-

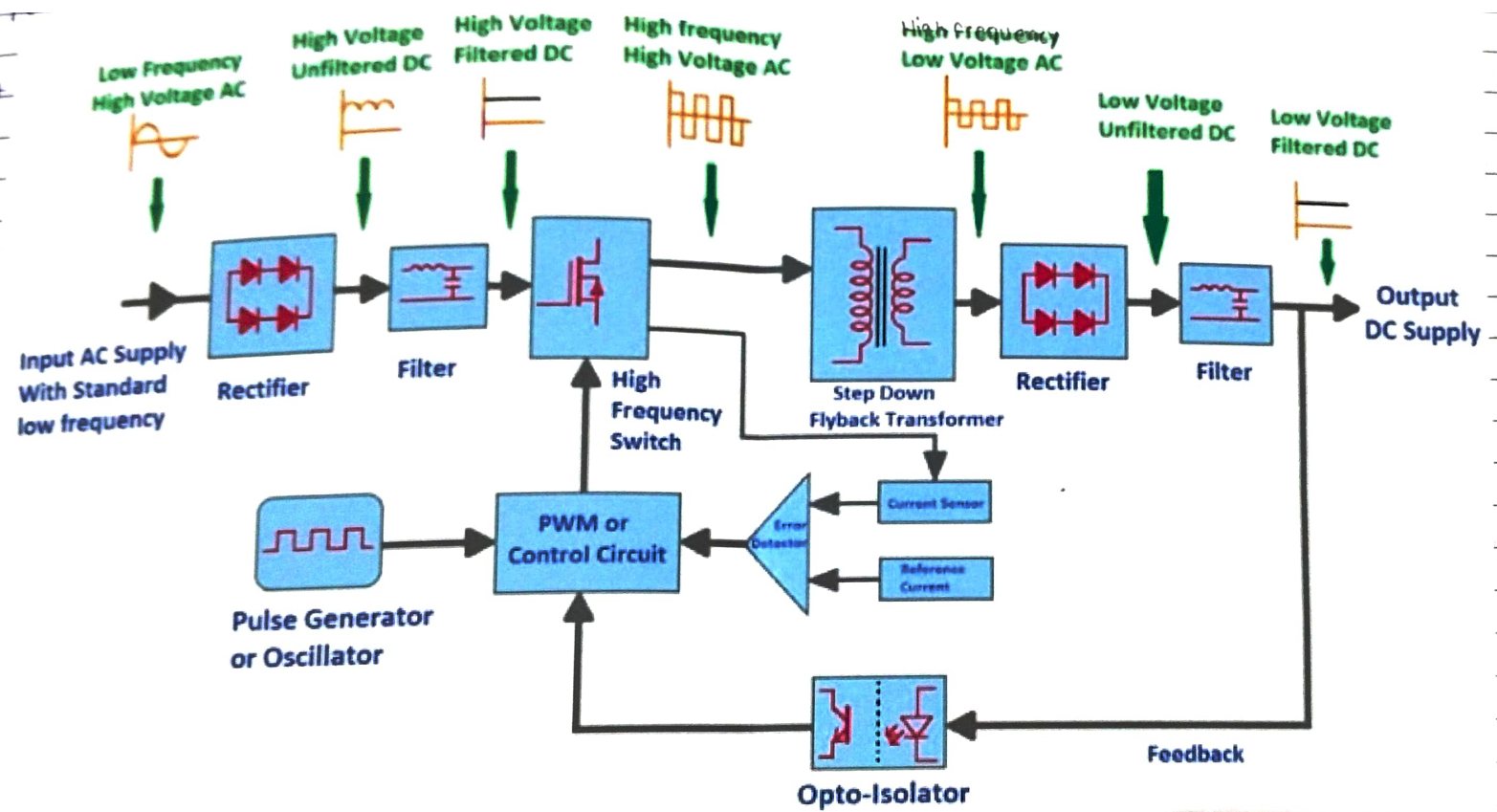
In a collage lab or home desktop computer provides immediate backup battery power during blackouts so you don't lose unsaved work or damage storage drives.

* Network server :-

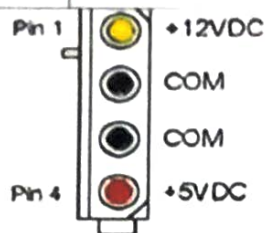
Prevents critical system shutdowns and data corruption by instantly switching to battery power during voltage sags or power outages.

Day 2.

2. Draw and label a diagram showing the main internal components of a typical snps unit found in desktop PC, including AC input, rectifier, transformer and DC output sections.



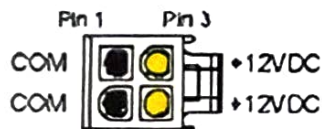
3. Take photos or find clear images online of the Power connectors: ATX 24-pin motherboard connector, SATA power connector, molex connector, and PCI 6 pin connector. For each write its name, the device it powers and one safety tip for connecting/disconnecting it.



Peripheral Power Connector

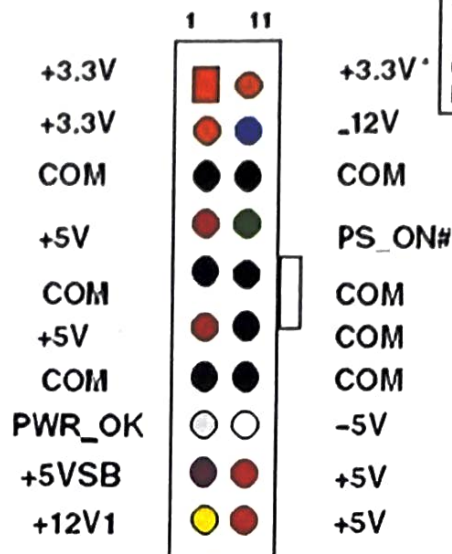


Floppy Drive Power Connector



+12V Power Connector

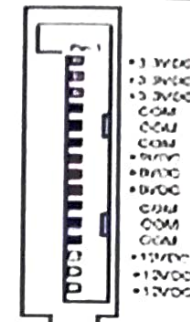
ATA Power Connector Reference



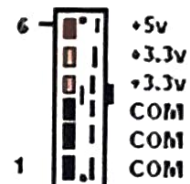
Main Power Connector

*May also be 3.3Vdc Sense (Brown)

SATA Connector Note:
The SATA connector uses only 5 wires for it's 15 connectors.



Serial ATA Connector



AUX Connector

+3.3V

+3.3V

COM

+5V

COM

+5V

COM

PWR_ON

+5VSB

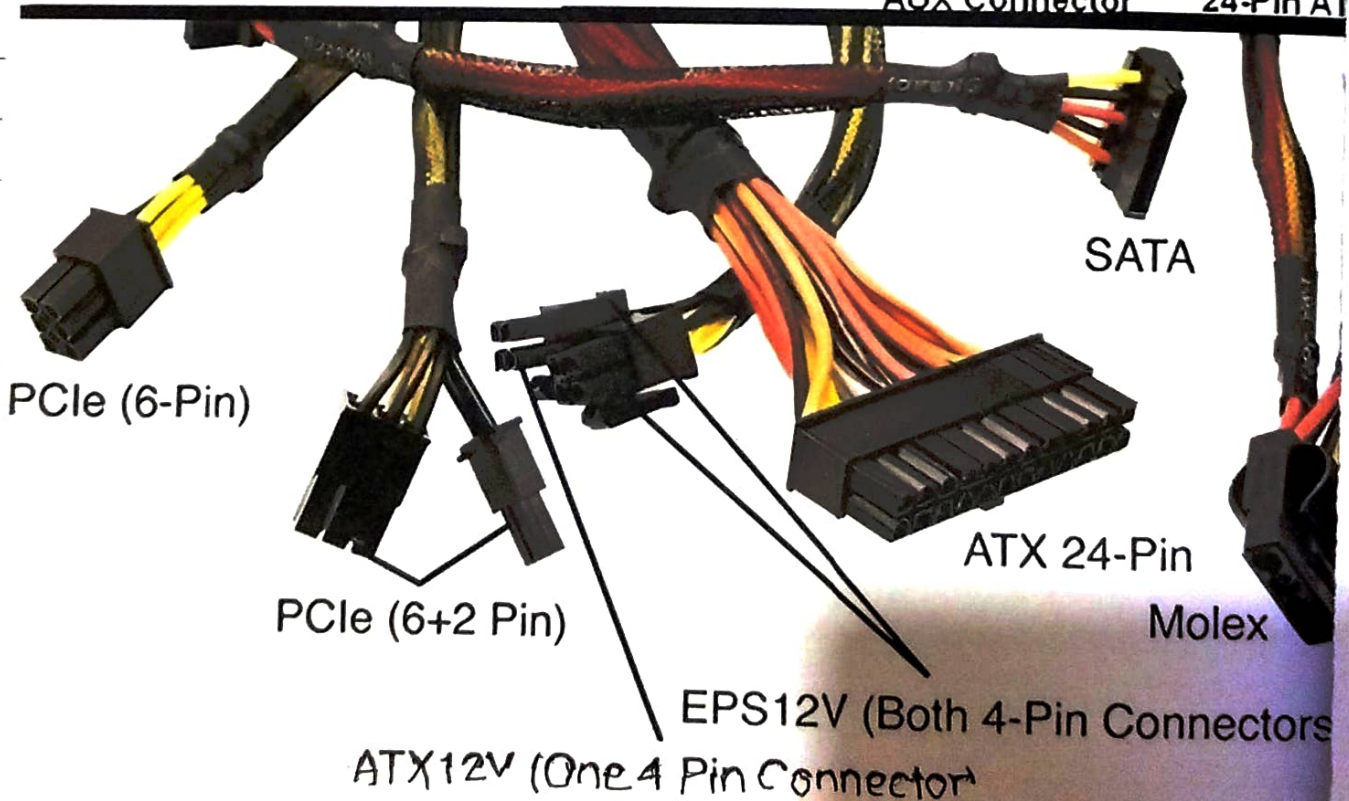
+12V1

+12V1

+3.3V

Main Power

24-Pin AT



Task- 4. imagine you are setting up a new gaming pc. List the steps you would follow to connect the Smps to the motherboard, graphics card, and storage devices, mentioning which specific cables & connectors are use for each.

→ Step 1 :- connect to motherboard :-

main power :-

plug the large 24 pin ATX cable into the right place on motherboard to power chipset & RAM.

cpu cable :-

plug 4 pin EPS cable into the right place on motherboard near processor to deliver power to CPU.

Step 2:- connect to Graphics card (GPU) &

Plug 6+2 pin cable for directly into the top power slot of your graphic card to supply heavy power needs.

Step 3:- connect to storage devices &

Plug a SATA power cable into SSD or HDD.